

Lecture 3

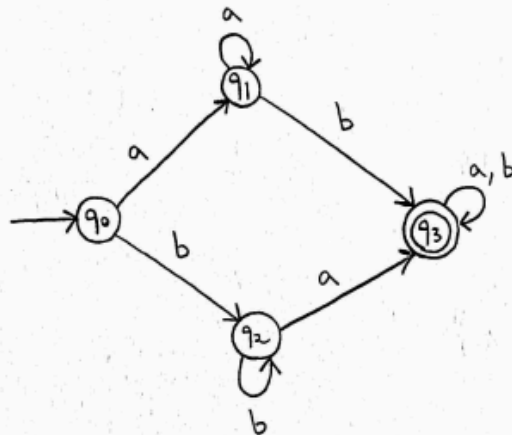
Complement

$L = \{w \mid w \in \{0, 1\}^* \text{ w contains neither the substring ab nor ba} \}$

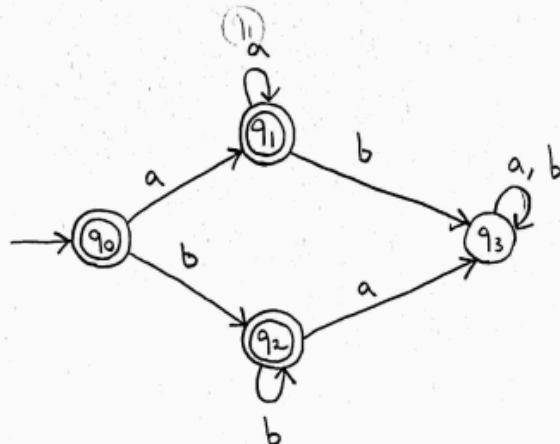
First complement the language:

$$\bar{L} = \{w \mid w \text{ contains either the substring ab or ba} \}$$

$\Rightarrow$ DFA for $\bar{L}$:



$\Rightarrow$ DFA for L : Accept states in $\bar{L}$ become reject states in L
Reject states in $\bar{L}$ become accept states in L



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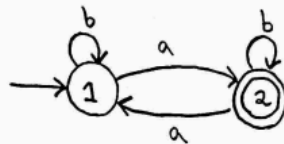
Union

$$L = \{ w \mid w \text{ has an odd number of } a\text{'s and ends with a } b \}$$

$$\Rightarrow L_1 = \{ w \mid w \text{ has an odd number of } a\text{'s} \}$$

$$\Rightarrow L_2 = \{ w \mid w \text{ ends with a } b \}$$

DFA for L_1 :



1. Set of states: $Q = \{1, 2\}$

2. Input alphabet: $\Sigma = \{a, b\}$

3. Transition function: $\delta: Q \times \Sigma \rightarrow Q$

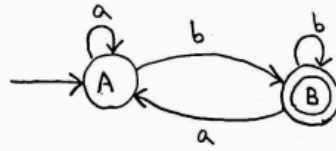
	a	b	
1	2	1	$\Rightarrow \delta(1, a) = 2, \delta(1, b) = 1$
2	1	2	$\Rightarrow \delta(2, a) = 1, \delta(2, b) = 2$

4. Start State: $q_0 = 1$

5. Set of Final States: $F = \{2\}$

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DFA for L_2 :



1. Set of states: $Q = \{A, B\}$
2. Input alphabet: $\Sigma = \{a, b\}$
3. Transition function: $\delta: Q \times \Sigma \rightarrow Q$

	a	b	
A	A	B	$\Rightarrow \delta(A, a) = A, \delta(A, b) = B$
B	A	B	$\Rightarrow \delta(B, a) = A, \delta(B, b) = B$

4. Start state: $q_0 = A$
5. Set of Final States: $F = \{B\}$

The two DFA L_1 and L_2 are existing separately.

Run the input **aaab** on the DFA for L_1 , the input will be consumed. This is a downside, since the DFA cannot rewind the input tape and then run the input in the DFA of L_2 . To solve this problem we need to run the DFA parallelly so both the conditions for the language are satisfied. We use a technique called cross product.

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DFA for L using L_1 and L_2 :

1. Set of states: $Q = \{1A, 1B, 2A, 2B\}$

2. Input alphabet: $\Sigma = \{a, b\}$

3. Transition function: For each $(r_1, r_2) \in Q$ and each $a \in \Sigma$, $\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$

$$\begin{aligned}\delta((1A), a) &= (\delta_1(1, a), \delta_2(A, a)) \\ &= (2, A)\end{aligned}$$

$$\begin{aligned}\delta((1A), b) &= (\delta_1(1, b), \delta_2(A, b)) \\ &= (1, B)\end{aligned}$$

$$\begin{aligned}\delta((1B), a) &= (\delta_1(1, a), \delta_2(B, a)) \\ &= (2, A)\end{aligned}$$

$$\begin{aligned}\delta((1B), b) &= (\delta_1(1, b), \delta_2(B, b)) \\ &= (1, B)\end{aligned}$$

$$\begin{aligned}\delta((2A), a) &= (\delta_1(2, a), \delta_2(A, a)) \\ &= (1, A)\end{aligned}$$

$$\begin{aligned}\delta((2A), b) &= (\delta_1(2, b), \delta_2(A, b)) \\ &= (2, B)\end{aligned}$$

$$\begin{aligned}\delta((2B), a) &= (\delta_1(2, a), \delta_2(B, a)) \\ &= (1, A)\end{aligned}$$

$$\begin{aligned}\delta((2B), b) &= (\delta_1(2, b), \delta_2(B, b)) \\ &= (2, B)\end{aligned}$$

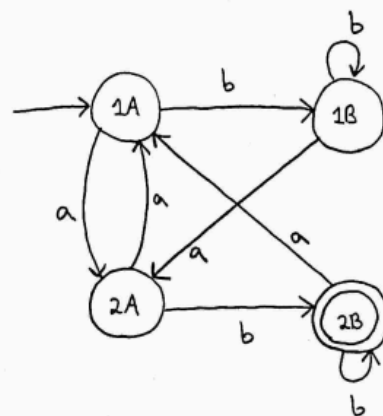
Transition Table for L :

	a	b
1A	2A	1B
1B	2A	1B
2A	1A	2B
2B	1A	2B

4. Start state: $q_0 = 1A$

5. Set of Final States: $F = \{2B\}$

DFA for L :



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AND in language

Set of final states $F = \{2B\}$

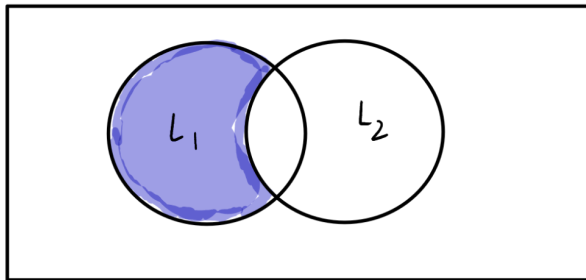
OR in language

Set of final states $F = \{1B, 2A, 2B\}$

XOR in language

Set of final states $F = \{1B, 2A\}$

What if the language said $L_1 - L_2 \rightarrow$ Difference



$$L_1 - L_2 = L_1 / L_2 \\ = L_1 \cap \bar{L}_2$$

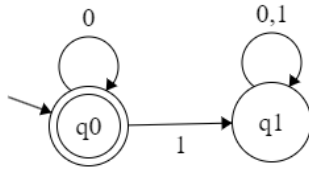
Here if we complement L_2 , the final state will be A.

$$\therefore F = \{2A\}$$

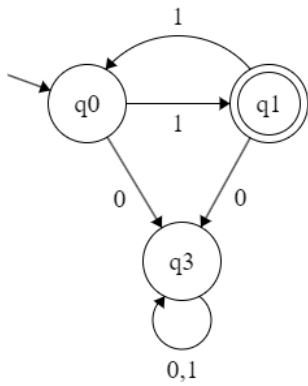
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Concatenation

$$L_1 = \{w \in \{0, 1\}^* \mid 0^n \text{ where } n \geq 0\}$$

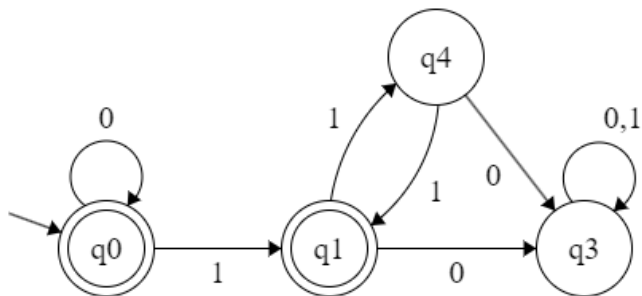


$$L_2 = \{w \in \{0, 1\}^* \mid 1^m \text{ where } m \text{ is odd}\}$$



$$L_3 = L_1 \circ L_2$$

In other words, the question means $L_3 = \{w \in \{0, 1\}^* \mid 0^n 1^m \text{ where } n \geq 0 \text{ and } m \text{ is odd}\}$

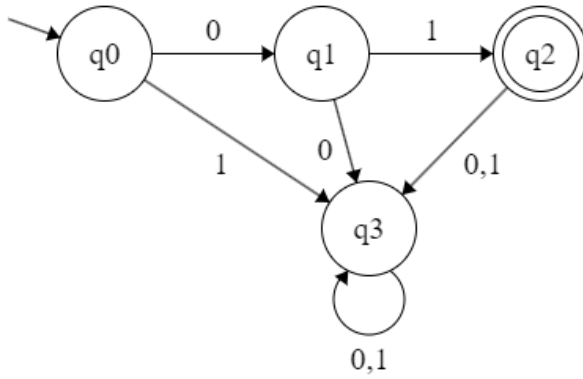


Note, there won't be a final state in q_0 .

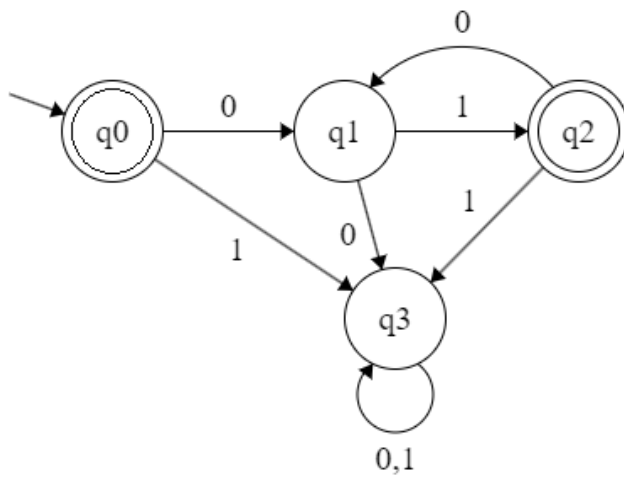
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Kleene star

$$L_4 = \{w \in \{0, 1\}^* \mid w \text{ is } 01\}$$



$$L_5 = L_4^*$$

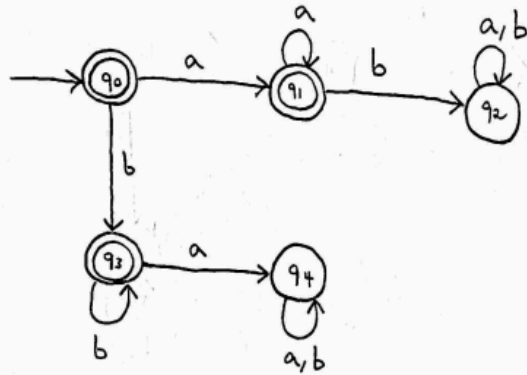


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$L_6 = \{w \in \{0, 1\}^* \mid w \text{ is any string not in } a^* \cup b^*\}$

$\Rightarrow \bar{L} = \{w \mid w \text{ is any string in } a^* \cup b^*\}$

$\Rightarrow$ DFA for $\bar{L}$:



$\Rightarrow$ DFA for L : Accept states in $\bar{L}$ become reject states in L
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